

Disproof of Conjecture 000000007669

earthking11

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Starting from (a, b) , write A for the move $a \mapsto aq$ and B for $b \mapsto bq$. At depth three the binary move histories are

$$AAA, AAB, ABA, ABB, BAA, BAB, BBA, BBB.$$

Thus, if the phrase “binary tree” means that histories are its nodes, the level-three node count is $2^3 = 8$.

There is a second possible reading: identify histories that produce the same parameter pair. But A and B commute, and a word containing i copies of A and j copies of B ends at (aq^i, bq^j) . The depth-three endpoint states are exactly

$$(3, 0), (2, 1), (1, 2), (0, 3),$$

where the pair records (i, j) . Hence the quotient count is 4.

The conjecture instead asserts

$$a_3 = \text{Fib}(3 + 2) = \text{Fib}(5) = 5.$$

It therefore fails under either natural meaning of “node”: $8 \neq 5$ before identification and $4 \neq 5$ after identification. This counterexample uses only the two moves supplied in the definition and does not depend on any analytic property of a Bailey identity.

The Lean file exhaustively lists the eight histories, maps them to exponent pairs, removes duplicates, and kernel-checks the counts 8, 4, and the Fibonacci value 5.